

# Success Metrics (KPIs) — counts-only, no-flow-by- construction

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**Document role.** This is the Volume 1 (Product & Requirements) nano-detail spec defining HelixVPN's product **KPIs** and the **telemetry that derives them** — strictly **counts and aggregates only**. The defining constraint of this document is that it is bound by v05-security/no-logging-as-code.md: a KPI we *cannot* measure without a per-user flow, a destination, or a timestamp-of-use is a KPI we **deliberately do not have** — and that absence is a *product feature*, not a gap. Every KPI names its counts-only metric source (citing v03-control-plane/svc-telemetry.md) and the privacy boundary it respects.

**Document is a SPEC.** It states *what we measure, from which counts-only series, against which target, within which privacy boundary*; it does not build the product. Every numeric target is a TARGET marked UNVERIFIED per §11.4.6 — a product KPI threshold is a business target, not a measured fact, until the instrumented series produces data.

**Evidence base.** Cross-references the Volume 1 overview (principles P7 no-logging, parity F14; differentiators X1-X5; the MVP DoD §10.2), the telemetry counts-only metric registry and its forbidden-label cardinality bound (v03-control-plane/svc-telemetry.md §3), the coordinator convergence SLO (v03-control-plane/svc-coordinator.md §7), and the no-logging invariant (v05-security/no-logging-as-code.md §2 the stored-vs-never-stored partition).

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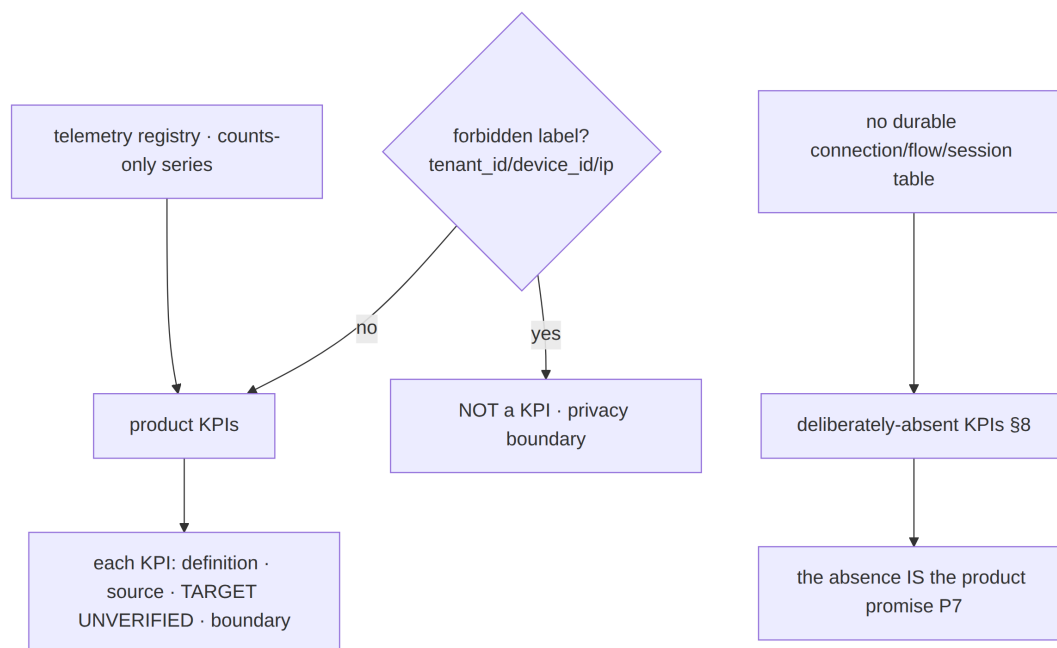
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## 1. The measurement charter (the privacy boundary is the design)

Most products measure success with per-user funnels, destination analytics, and session timelines. HelixVPN **cannot** — by construction (P7 / S6): no durable connection/traffic/flow/session table exists, the event bus carries no src/dst/bytes/dns field, and / metrics forbids any tenant\_id/device\_id/ip label. Every KPI in this document is therefore derived from **aggregate counters, SLO histograms, and bounded gauges** — never from a per-user trail.

The charter, stated as three rules every KPI obeys:

1. **Counts and aggregates only.** A KPI's source is a Prometheus counter, histogram, or bounded gauge from the telemetry registry (svc-telemetry.md §3), or an aggregate roll-up of those — never a per-user row (there is none).
2. **No forbidden label.** A KPI may slice by a *bounded* label (action, event\_type, stream, build version) but **never** by tenant\_id, device\_id, overlay\_ip, or any traffic/PII-shaped key — that label set is forbidden by the cardinality guard (svc-telemetry.md §3.1).
3. **The absence is a feature.** Where a conventional KPI would need a per-user flow / destination / timestamp-of-use, HelixVPN has no such KPI **on purpose** (§8). "We can't tell you which sites a user visited or when they were online" is the product promise, surfaced as a measurement boundary.



**§11.4.6 honesty.** Every numeric target below is a *business TARGET* marked UNVERIFIED — KPI thresholds are goals, not measurements. A KPI is "met" only when its named counts-only series produces data showing it; the threshold written here is never asserted as already achieved.

## 2. What we can measure vs what we deliberately cannot

The closed partition, drawn directly from the no-logging stored-vs-never-stored model (no-logging-as-code.md §2) projected onto the KPI layer:

| Measurable (counts/aggregates exist)                         | Deliberately NOT measurable (no data exists)                                            |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Total devices enrolled (registry count)                      | Which destinations any device reached                                                   |
| Devices currently online (bounded presence gauge, aggregate) | When a specific user was online (no session timeline; last_seen_at coarse $\geq 5$ min) |
| Total connectors / advertised networks (topology count)      | Which user used which network at which time                                             |
| Handshakes total, by transport-kind (aggregate counter)      | Per-user transport history / per-user flow                                              |
| Convergence/revoke SLO percentiles (histograms)              | Per-user latency or per-destination latency                                             |

| Measurable (counts/<br>aggregates exist)            | Deliberately NOT measurable<br>(no data exists)                        |
|-----------------------------------------------------|------------------------------------------------------------------------|
| Aggregate bytes-total (no<br>per-peer, no per-user) | Per-user / per-destination byte<br>counts                              |
| Audit control-action counts,<br>by action (bounded) | Any traffic event (none is audited<br>— none exists)                   |
| Ladder escalation<br>outcomes, aggregate by<br>rung | Which user escalated on which<br>network (no SSID, no endpoint,<br>I5) |

The right column is not "unimplemented" — it is **architecturally impossible** to measure because the data is never written (S6). This is the §11.4.112 structural property turned into a product virtue: the metric cannot be reconstructed because there is nothing to reconstruct it from.

### 3. Adoption & reach KPIs (counts-only)

| KPI                                       | Definition                                                                | Counts-only source                                                                                        |
|-------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>K-ADOPT-1<br/>Devices<br/>enrolled</b> | total devices<br>ever enrolled<br>per tenant<br>(and global<br>aggregate) | registry row count; device.enrolled audit counter<br>(helix_audit_events_total{action="device.enrolled"}) |
| <b>K-ADOPT-2<br/>Active<br/>devices</b>   | devices<br>currently<br>online                                            | helix_presence_online_devices (bounded gauge,<br>reconciled from SCARD)                                   |
| <b>K-ADOPT-3<br/>Networks<br/>joined</b>  | total<br>connectors<br>attached +<br>advertised<br>prefixes (the          | registry/topology counts (advertised_prefixes rows)                                                       |

| KPI                                  | Definition                                                                 | Counts-only source                                                                    |
|--------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                                      | X1 multi-network differentiator)                                           |                                                                                       |
| <b>K-ADOPT-4 Multi-network reach</b> | distribution of networks-reachable-per-user (the 1 user → N nets headline) | derived aggregate from compiled-policy VisibleTo cardinality (counts, not identities) |
| <b>K-ADOPT-5 Apps in use</b>         | which of the three app classes (Access/Connector/Console) are active       | helix_build_info + open-stream class aggregate                                        |

**K-ADOPT-2 boundary (§11.4.6).** "Active devices" is an *instantaneous* aggregate count derived from the ephemeral presence set — it deliberately cannot answer "was device X online yesterday at 14:00" because presence is TTL'd and last\_seen\_at is coarsened to  $\geq 5$  min and carries no destination (no-logging §2.2). The point-in-time count is the metric; the timeline is the deliberately-absent one (§8).

## 4. Reliability & performance KPIs (SLO-derived)

These KPIs are roll-ups of the SLO histograms the coordinator + telemetry already expose (svc-coordinator.md §7, svc-telemetry.md §3.2). They measure *system* quality, never *user* behaviour.

| KPI                                       | Definition                                   | Counts-only source                | Ta                   |
|-------------------------------------------|----------------------------------------------|-----------------------------------|----------------------|
| <b>K-REL-1 Convergence SLO attainment</b> | % of time p99(helix_reconcile_seconds) < 1 s | helix_reconcile_seconds histogram | <b>TA p9 (Do UNV</b> |

| KPI                                           | Definition                                         | Counts-only source                                   | Target                                      |
|-----------------------------------------------|----------------------------------------------------|------------------------------------------------------|---------------------------------------------|
| <b>K-REL-2<br/>Revoke SLO attainment</b>      | % of revokes enforced < 1 s                        | helix_revoke_enforce_seconds histogram               | TA<br>< 1 s<br>UNV                          |
| <b>K-REL-3<br/>Availability (fail-static)</b> | tunnel-drops attributable to control-plane outages | chaos-test outcome + helix_presence_backend_up gauge | TA<br>0 c<br>on<br>com<br>pla<br>out<br>(N) |
| <b>K-REL-4<br/>Event-pipeline health</b>      | DLQ accumulation + stream backlog                  | helix_events_dlq_total, helix_stream_pending_entries | TA<br>DL<br>bac<br>bot<br>—<br>UNV          |
| <b>K-REL-5 No-leak resource posture</b>       | coordinator memory slope over soak                 | process_resident_memory_bytes deriv                  | TA<br>slo<br>ove<br>(N)                     |
| <b>K-REL-6<br/>Handshake success rate</b>     | successful WG handshakes / attempts (aggregate)    | aggregate handshake counter (edge /metrics)          | TA<br>—<br>UNV                              |

**K-REL-1 is the product's flagship quality KPI.** It is the parity-grade real-time promise (P4) made into a falsifiable series. It carries **no** tenant or device label (§3.1 cardinality guard), so it measures the *system's* convergence, never any user's activity.

## 5. Privacy-integrity KPIs (the guarantee is itself a KPI)

The most important KPIs invert the usual relationship: the *strength of the privacy guarantee* is measured, and any erosion is a release-blocking failure.

| KPI                                                        | Definition                                                                            | Counts-only source                                                     | Target                                                                         | P                                |
|------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------|
| <b>K-PRIV-1<br/>No-logging<br/>gate green</b>              | schema-lint exit-0<br>against<br>migrations<br>AND the<br>deployed<br>DB              | schemalint exit code (pre-build<br>+ post-deploy runtime<br>signature) | <b>TARGET 100</b><br>% green<br>(NFR-300/308)                                  | m<br>th<br>pr<br>ak<br>st<br>no  |
| <b>K-PRIV-2<br/>Lint not a<br/>tautology</b>               | paired<br>§1.1<br>mutation:<br>planted<br>flows table<br>→ lint FAIL                  | meta-test result                                                       | <b>TARGET</b><br>golden-bad<br>FAILs every<br>build<br>(NFR-301)               | m                                |
| <b>K-PRIV-3<br/>Bus payload<br/>clean</b>                  | zero<br>traffic-<br>shaped<br>field in<br>any event<br>payload                        | payload-lint result                                                    | <b>TARGET 0</b><br>forbidden<br>fields<br>(NFR-302)                            | m                                |
| <b>K-PRIV-4<br/>Audit<br/>control-only</b>                 | audit rows<br>are<br>control<br>actions<br>only;<br>meta-<br>shape<br>guard<br>active | helix_audit_events_total{action}<br>(closed vocab) + reject counter    | <b>TARGET 0</b><br>traffic-shaped<br>meta rejected-<br>into-store<br>(NFR-303) | bo<br>ac<br>la                   |
| <b>K-PRIV-5<br/>Metric<br/>cardinality<br/>bound</b>       | every<br>collector's<br>label set $\subseteq$<br>allow-list                           | label-audit unit test                                                  | <b>TARGET 100</b><br>% pass<br>(NFR-306)                                       | th<br>en<br>th<br>bo<br>it<br>un |
| <b>K-PRIV-6<br/>Anonymous-<br/>enroll<br/>availability</b> | tenants<br>can enroll<br>devices<br>with no<br>email/SSO                              | integration test result                                                | <b>TARGET</b><br>available<br>(NFR-307,<br>F15)                                | th<br>th<br>th<br>bo<br>us<br>m  |

**K-PRIV-5 is reflexive.** It measures that the KPI layer itself never grew a forbidden label — the success metric for "we did not accidentally start tracking users" is itself a counts-

only, label-bounded check. A green K-PRIV-1..6 is the captured-evidence proof (§11.4.69) that the privacy promise is mechanically intact, not merely asserted.

## 6. Censorship-resistance KPIs (aggregate ladder outcomes)

The transport ladder records **aggregate, no-per-user** escalation outcomes (I5) — exactly enough to know "is obfuscation working in aggregate" without ever knowing *who* escalated *where*.

| KPI                                           | Definition                                                      | Counts-only source                                | Target                                                             | Privacy boundary                                                        |
|-----------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>K-CENS-1 Obfuscation success rate</b>      | % of connect attempts that reach a working rung (aggregate)     | aggregate ladder-outcome counter by terminal rung | <b>TARGET</b> high — UNVERIFIED                                    | aggregate by rung-kind; no SSID, no endpoint, no per-user (I5)          |
| <b>K-CENS-2 DPI-survival throughput</b>       | obfuscated throughput as % of plain (the Phase-0 G2 bar)        | benchmark capture (netns DPI rig)                 | <b>TARGET</b> $\geq 50\%$ of plain (NFR-002) — UNVERIFIED until G2 | benchmark, not production user data                                     |
| <b>K-CENS-3 Escalation-depth distribution</b> | how far up the ladder attempts go, in aggregate                 | histogram of terminal-rung index (counts)         | <b>TARGET</b> — UNVERIFIED                                         | aggregate distribution; no per-network identity (fingerprint hash only) |
| <b>K-CENS-4 Time-to-working-rung</b>          | aggregate latency to a successful WG handshake under escalation | aggregate timing histogram                        | <b>TARGET</b> bounded (NFR-006) — UNVERIFIED                       | aggregate timing; no per-user trace                                     |

**K-CENS boundary (§11.4.6).** The ladder's per-network memory persists a fingerprint *hash* + rung index — so K-CENS-3 can report "X% of networks needed MASQUE" in



aggregate but **cannot** name a network, an SSID, or a user. The escalation event trace is the captured evidence (L-I9) and carries no destination.

## 7. Self-host adoption KPIs (the differentiator)

Self-hostability (P6 / X3) is the product's reason to exist; its success is measured by deployment-shape counts, never by reaching into deployed instances (a self-hoster's instance is theirs — HelixVPN cannot and must not phone home per-user data).

| KPI                                         | Definition                                                                          | Counts-only source                       | Target                                                 | Privacy boundary                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------|
| <b>K-SELF-1 Self-host bootstrap success</b> | a clean VPS reaches a working deployment via <code>helixvpncctl init</code> (DoD-1) | bootstrap test / Challenge outcome       | <b>TARGET 100 %</b> on the supported path — UNVERIFIED | test outcome; no telemetry from operator instances by default    |
| <b>K-SELF-2 8-criteria DoD attainment</b>   | how many of the MVP DoD criteria pass on a clean self-host                          | DoD acceptance-suite result (8 criteria) | <b>TARGET 8/8</b> — UNVERIFIED                         | acceptance evidence, not user data                               |
| <b>K-SELF-3 Single-pod resource fit</b>     | the self-host pod runs within homelab resource bounds + the \$12.6 60 % ceiling     | host resource sampler (§11.4.24)         | <b>TARGET</b> within budget — UNVERIFIED               | host-side metric on the <i>test</i> host, not operator instances |
| <b>K-SELF-4 Image-set parity</b>            | the same OCI image set serves self-host and fleet (no code fork)                    | build/artifact check                     | <b>TARGET</b> one image set — UNVERIFIED               | artifact metadata, not user data                                 |

**No phone-home by default (§11.4.6 / P7).** A self-hosted instance does not emit per-user telemetry to any HelixVPN-operated endpoint; K-SELF KPIs are measured from the *project's own* acceptance suite and test hosts, never by harvesting operator deployments. A managed SKU may collect *aggregate* operational counters from its *own* fleet under the same counts-only boundary — never per-user flows.

## 8. The deliberately-absent metrics (and why each absence is a feature)

This is the document's thesis table: the conventional KPIs HelixVPN **deliberately does not have**, each paired with the product promise its absence encodes.

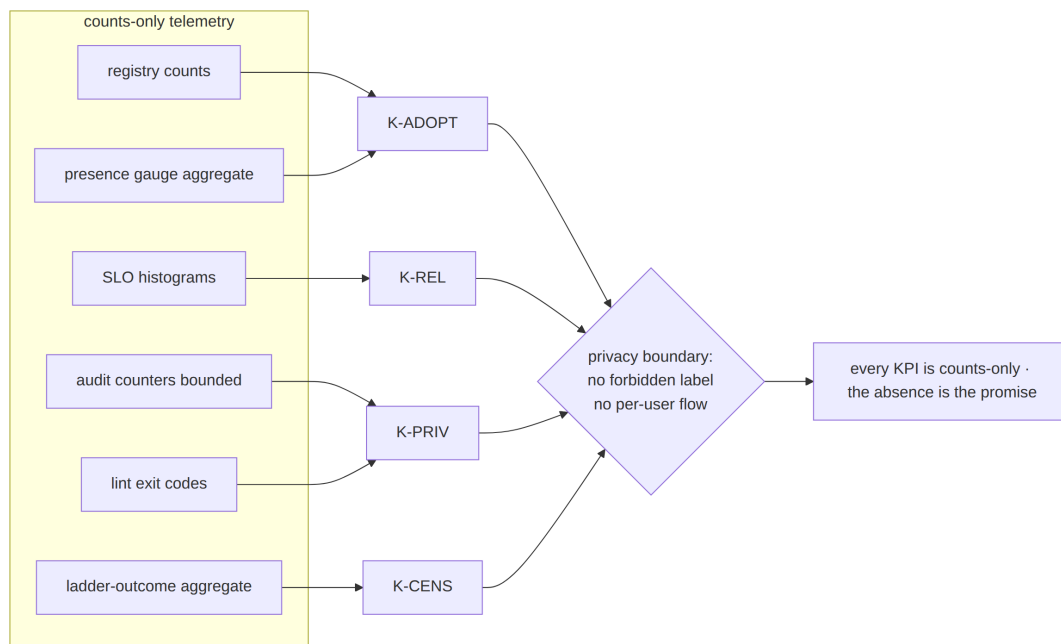
| Conventional KPI we do NOT have                  | Why it cannot exist                                                                      | The feature the absence encodes                          |
|--------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Destinations per user / "top sites"</b>       | no durable destination column anywhere (S6)                                              | "we never know where you go" — the core privacy promise  |
| <b>Per-user session timeline / "usage hours"</b> | presence is ephemeral TTL; <code>last_seen_at</code> coarse $\geq 5$ min, no destination | "we can't reconstruct when you were online"              |
| <b>Per-user bandwidth / data-cap analytics</b>   | no per-user/per-peer byte counts; only aggregate bytes-total                             | "we don't meter or profile your traffic"                 |
| <b>Per-user transport history</b>                | ladder memory is a fingerprint hash + rung index, no per-user link (I5)                  | "we don't track which networks you fought censorship on" |
| <b>DNS query analytics / "popular domains"</b>   | no DNS query log table (S6)                                                              | "we never log your DNS" (F13)                            |
| <b>Funnel/cohort analysis keyed to a person</b>  | no <code>tenant_id/device_id</code> label on metrics; no PII-linked event                | "you can be fully anonymous (F15)"                       |
| <b>Per-tenant breakdown on public /metrics</b>   | forbidden labels; per-tenant only behind                                                 | "your population is not leaked to a scraper"             |

| Conventional<br>KPI we do NOT<br>have | Why it cannot exist         | The feature the<br>absence encodes |
|---------------------------------------|-----------------------------|------------------------------------|
|                                       | authenticated /v1/<br>stats |                                    |

**§11.4.112 framing.** Each absence is a *structural impossibility*, not a backlog item: the data is never written, so the metric is unreconstructable. Reintroducing any of these would require adding a forbidden table/column/label that the schema-lint + payload-lint + cardinality guard would *fail the build* over (K-PRIV-1..5). The absence is therefore self-enforcing.

## 9. KPI → metric source → privacy-boundary traceability

| KPI family                 | Counts-only<br>source                           | Principle /<br>parity   | Privacy guard<br>that bounds it                                |
|----------------------------|-------------------------------------------------|-------------------------|----------------------------------------------------------------|
| Adoption (K-ADOPT)         | registry counts, presence gauge, audit counters | X1 multi-network        | no tenant_id/device_id label; per-tenant only behind /v1/stats |
| Reliability (K-REL)        | SLO histograms, process gauges                  | P4 push, P1 fail-static | no per-user/tenant label on histograms                         |
| Privacy-integrity (K-PRIV) | lint exit codes, label-audit tests              | P7 no-logging, F15      | the KPIs <i>are</i> the boundary enforcers                     |
| Censorship (K-CENS)        | aggregate ladder-outcome counters               | F2/F6/F7                | fingerprint hash only; no SSID/endpoint/user (I5)              |
| Self-host (K-SELF)         | acceptance-suite + build checks                 | P6/X3                   | no phone-home; test-host metrics only                          |



**Closing honesty (§11.4.6).** No target in §3–§7 is a measured result; each is a business TARGET marked UNVERIFIED and met only when its named counts-only series produces data. And by design, no KPI anywhere in this document can be computed from a per-user flow, a destination, or a timestamp-of-use — because that data never exists (P7 / S6). The measurement boundary *is* the product.

## Sources verified

- docs/research/mvp/final/00-product-scope-and-principles.md — principle P7 (no-logging by construction §8), parity F13/F14/F15 (§9), differentiators X1/X3 (§9), the 8-criteria MVP DoD (§10.2).
- docs/research/mvp/final/v03-control-plane/svc-telemetry.md §3 — the complete counts-only Prometheus registry (helix\_reconcile\_seconds, helix\_revoke\_enforce\_seconds, helix\_events\_dlq\_total, helix\_stream\_pending\_entries, helix\_presence\_online\_devices, helix\_audit\_events\_total{action}, helix\_build\_info, process\_resident\_memory\_bytes); §3.1 the forbidden-label cardinality bound (no tenant\_id/device\_id/ip label); §5 ephemeral presence; the per-tenant /v1/stats vs public /metrics split.
- docs/research/mvp/final/v03-control-plane/svc-coordinator.md §7 — the convergence < 1 s and revoke < 1 s SLO histograms feeding K-REL.
- docs/research/mvp/final/v05-security/no-logging-as-code.md — the stored-vs-never-stored partition (§2), coarse last\_seen\_at (§2.2), the schema-lint + payload-lint + meta-shape guard +

paired §1.1 mutation feeding K-PRIV, the deliberately-absent data classes (§8) and the structural-impossibility framing.

- docs/research/mvp/final/v02-data-plane/transport-selection-ladder.md — the aggregate, no-per-user ladder telemetry (I5) and per-network fingerprint-hash memory feeding K-CENS.

*Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — every KPI target marked UNVERIFIED, never asserted as measured), §11.4.69 (KPIs derived only from captured counts-only series), §11.4.112 (the deliberately-absent metrics framed as structural impossibilities, not backlog), the no-logging covenant (P7 / S6) bounding what any KPI may measure, §11.4.65/ §11.4.153 (HTML+PDF[+DOCX] exports follow in refinement).*